

Arabic LLM Bake-off Analytics

Report generated from analytics/llm_outputs

Models OK: 4 / 7

Best robot realtime: Qwen3-4B-Instruct-2507

Lowest TTFT: Nile-Chat-4B

Highest throughput: Qwen3-4B-Instruct-2507

Lowest VRAM: Qwen3-8B

Note: single-prompt pilot (A001 only).

Use ranks for latency/VRAM screening, not final quality selection.

Notes

- Automated scores cover latency/throughput/resources only.
- Manually review Egyptian / MSA / code-switch response quality in llm_outputs/responses/.
- For robot UX, prioritize low TTFT even if peak tok/s is slightly lower.
- Disable thinking modes where available to reduce TTFT further.

Recommended picks

Best for robot realtime

Model: Qwen3-4B-Instruct-2507

Why: Best composite of TTFT + tok/s + VRAM + load for conversational turns.

Metrics: score_robot_realtime=92.53, avg_first_token_seconds=0.618, avg_tokens_per_second=11.98, peak_vram_mb=8430.2

Lowest TTFT

Model: Nile-Chat-4B

Why: Fastest average time-to-first-token (best turn-taking feel).

Metrics: avg_first_token_seconds=0.538

Highest throughput

Model: Qwen3-4B-Instruct-2507

Why: Highest average tokens/second.

Metrics: avg_tokens_per_second=11.98

Lowest VRAM

Model: Qwen3-8B

Why: Lowest peak VRAM — better for small GPUs / co-residency.

Metrics: peak_vram_mb=6722.2

Best balanced

Model: Qwen3-4B-Instruct-2507

Why: Balanced latency/throughput/VRAM tradeoff.

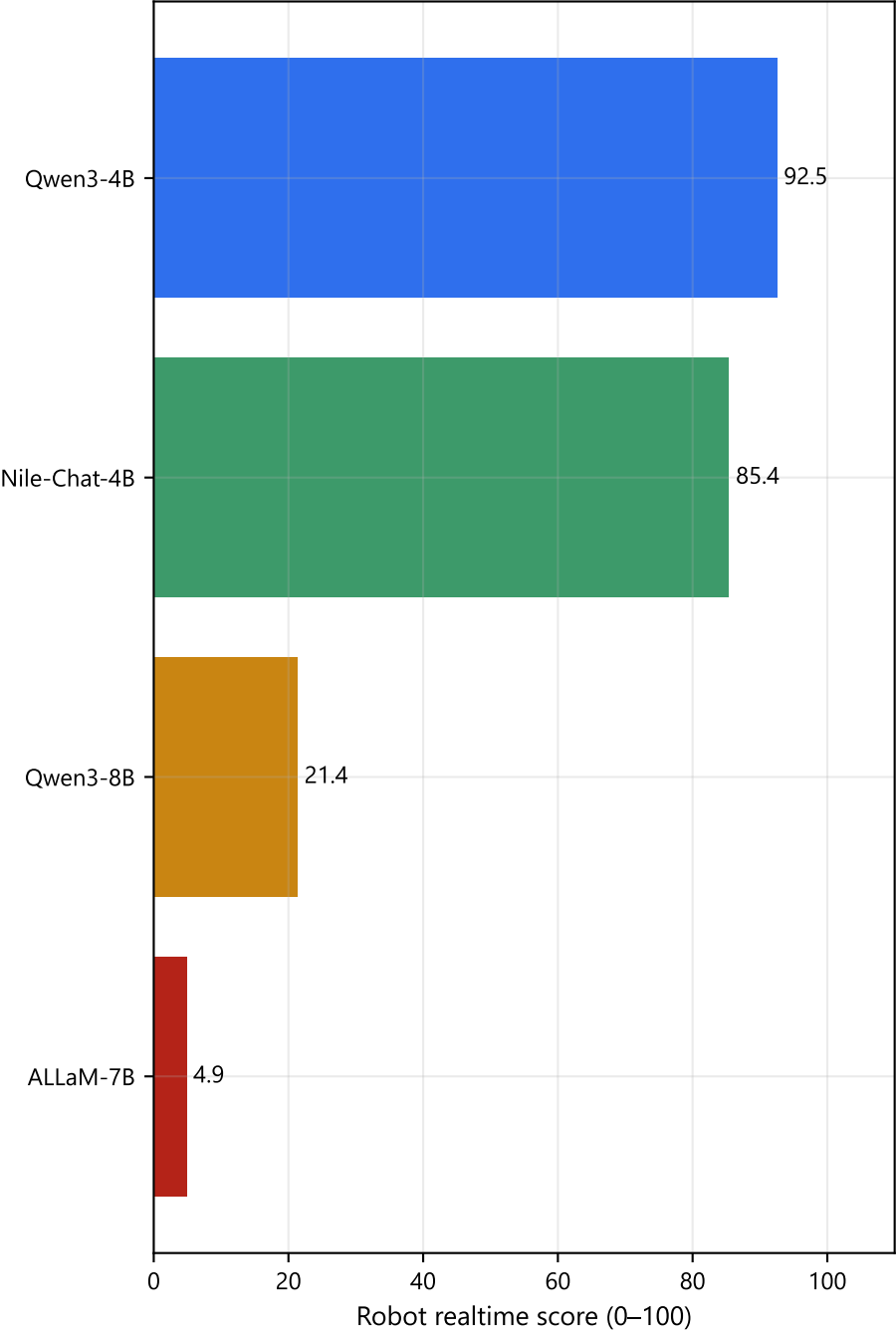
Metrics: score_balanced=91.59

Category specialists

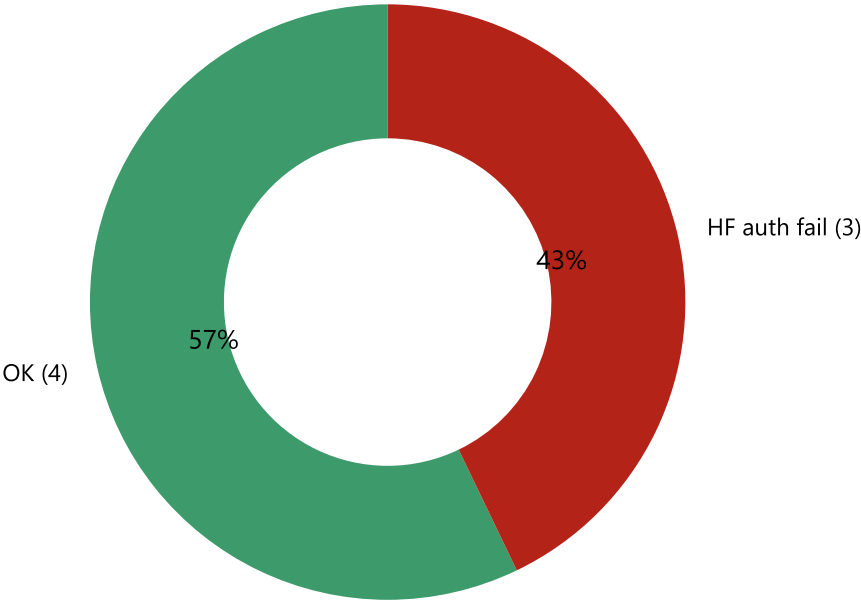
- egyptian_arabic: Nile-Chat-4B (TTFT=0.538, tok/s=8.9)

Robot score & run outcomes

Robot realtime score by model

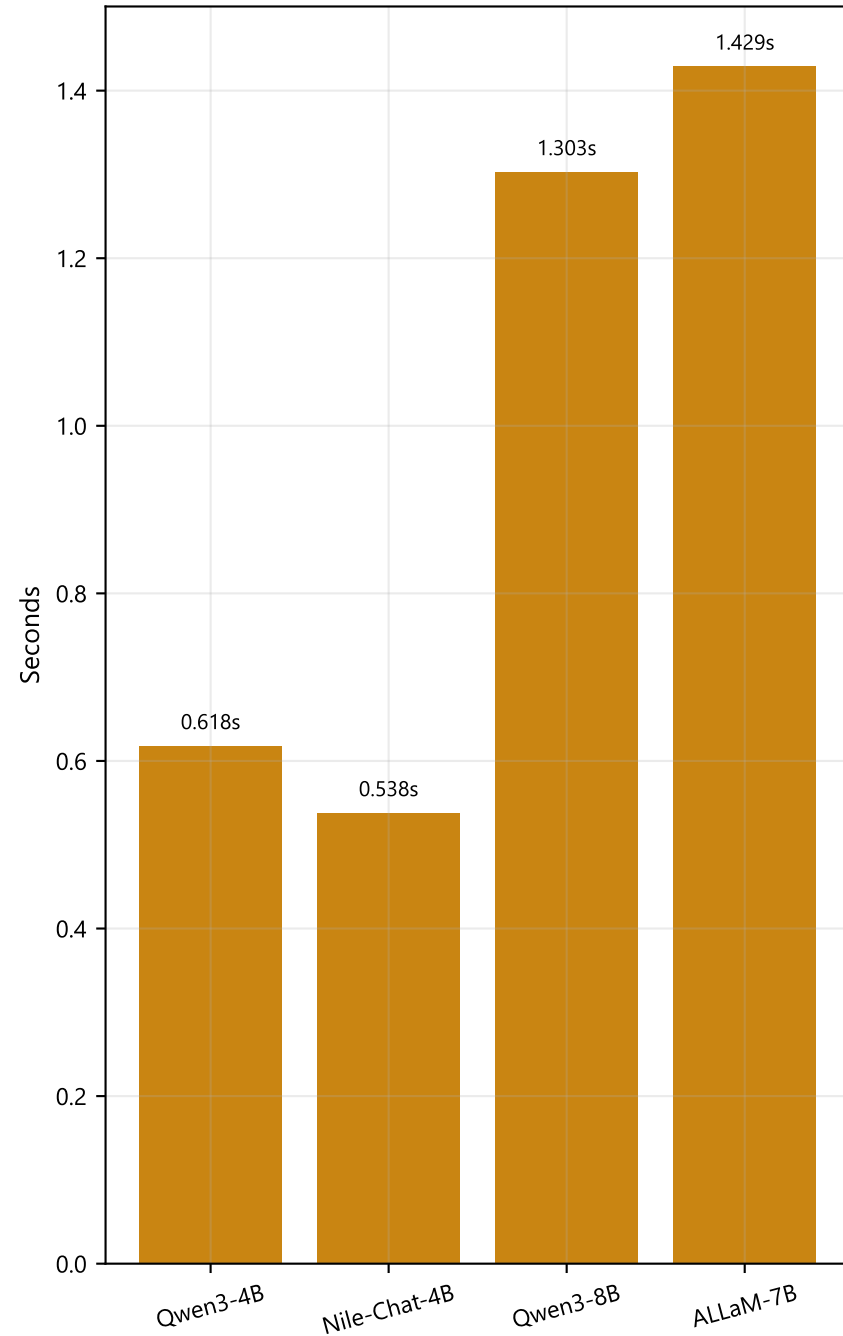


Run outcomes (7 models × 1 prompt)

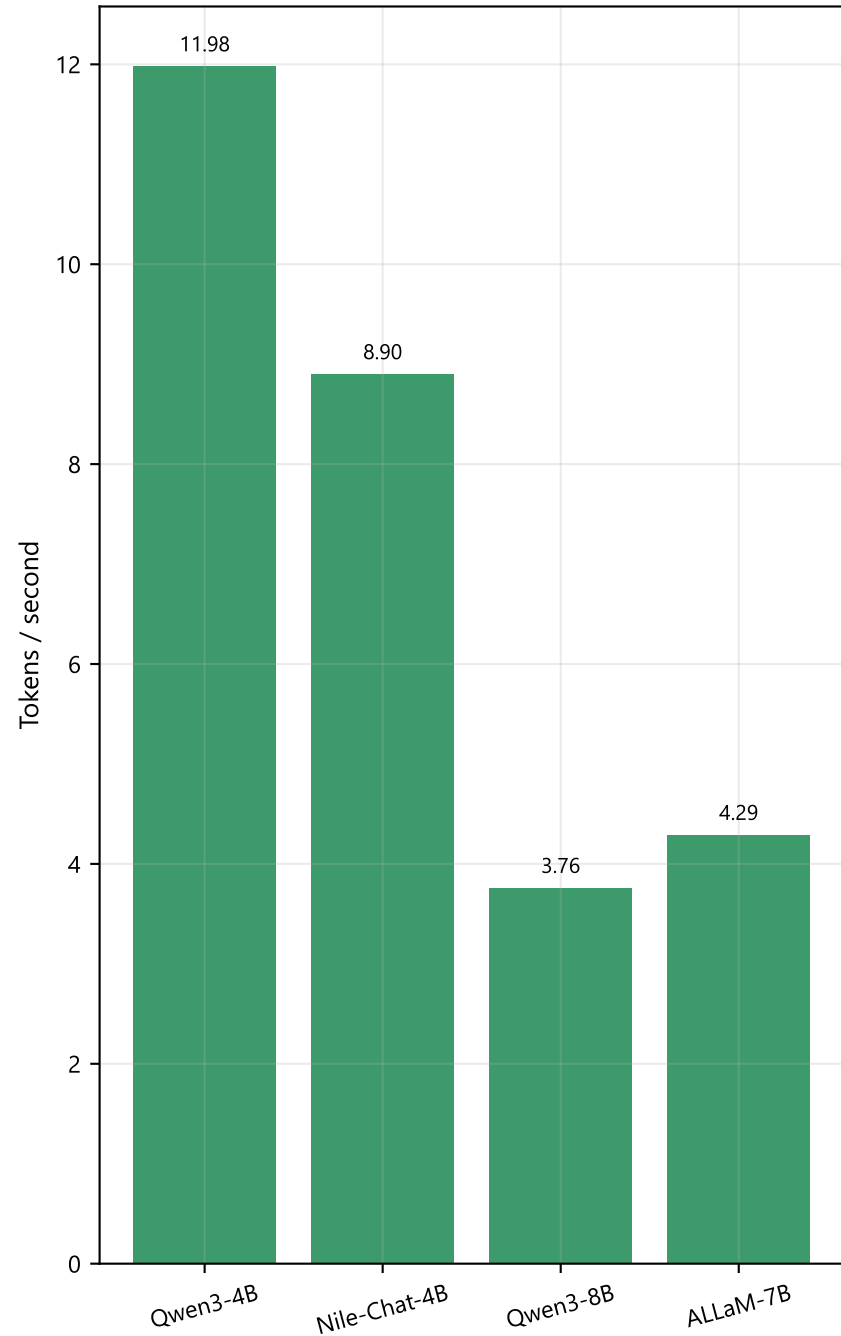


Latency & throughput

Time to first token (lower is better)

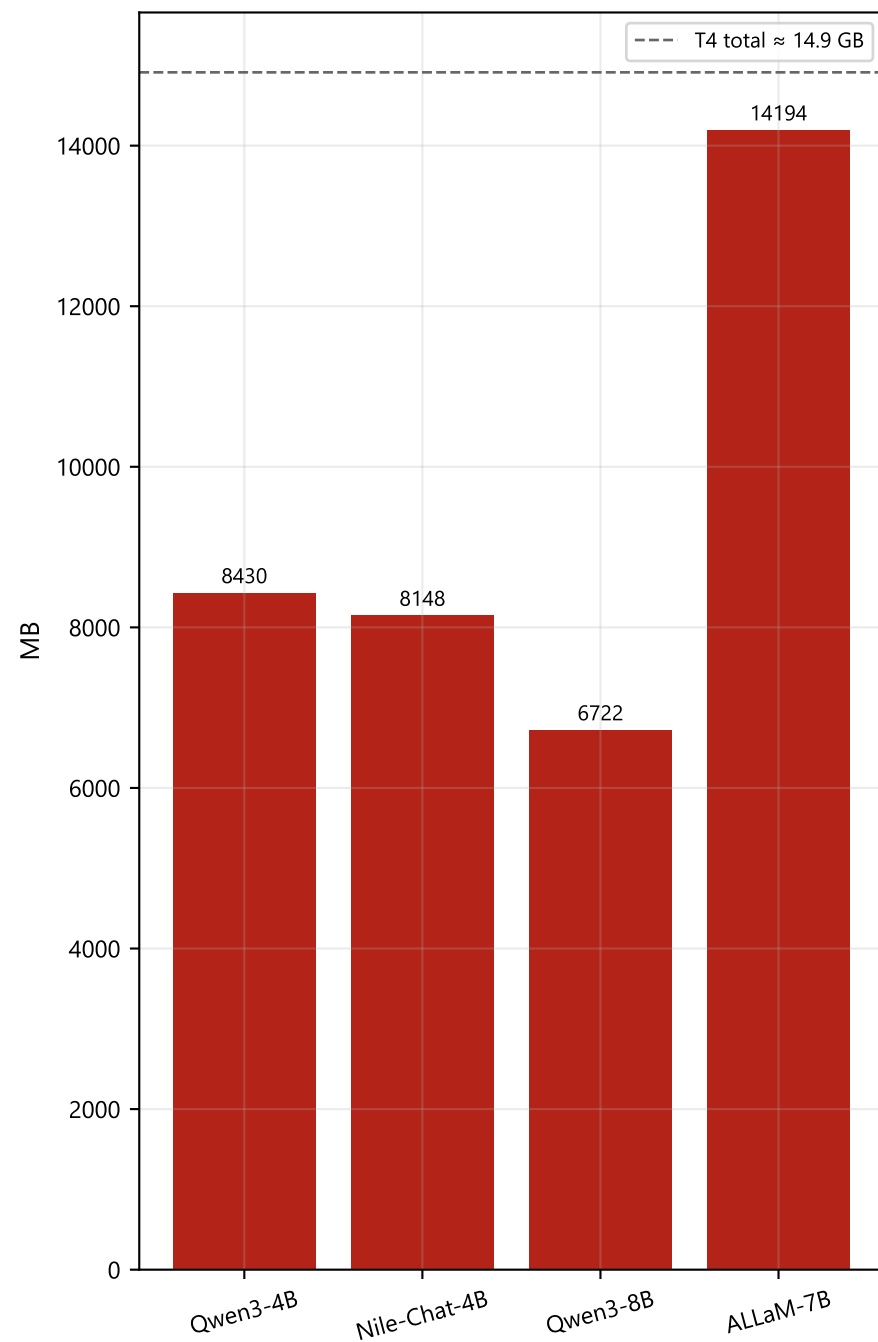


Generation throughput (higher is better)

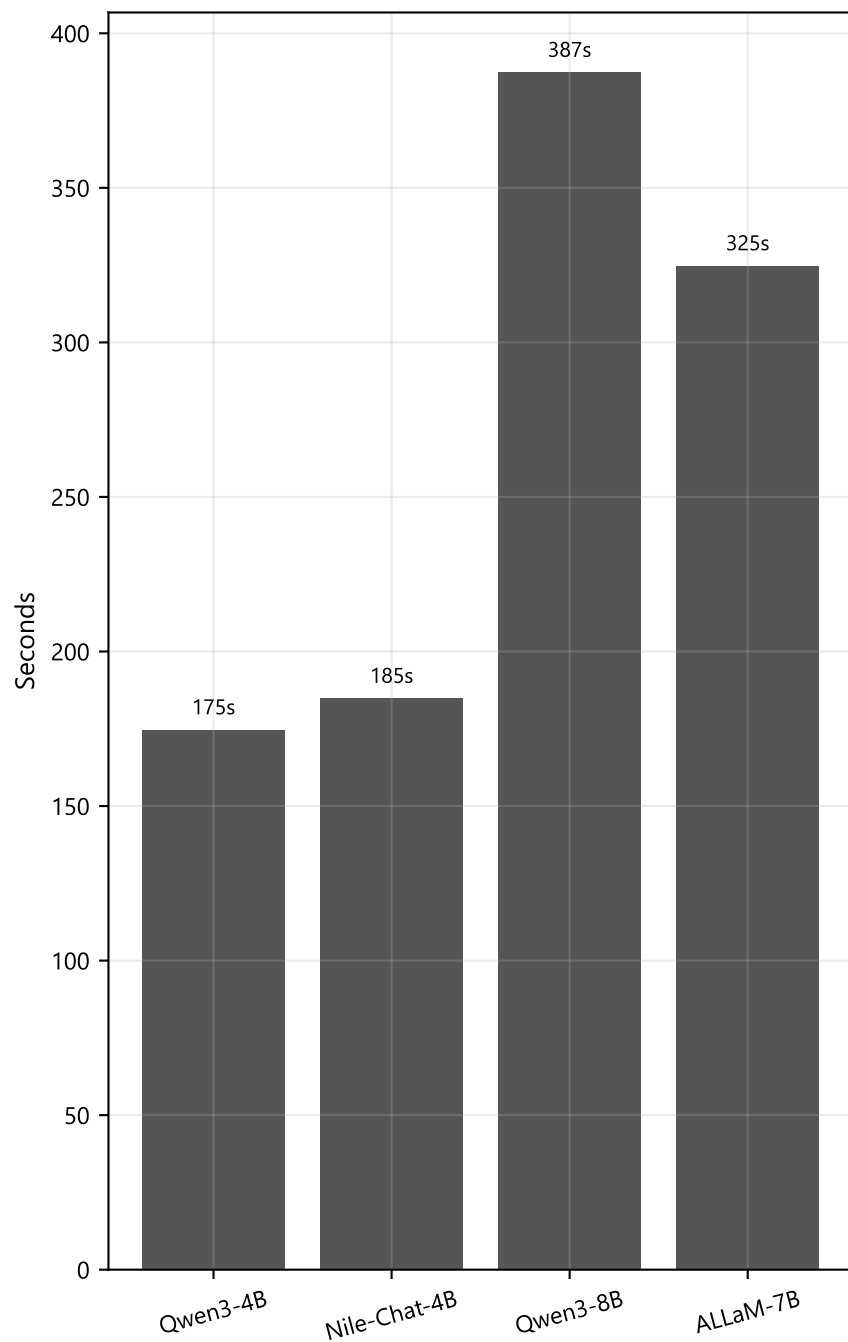


VRAM & model load time

Peak VRAM on Tesla T4

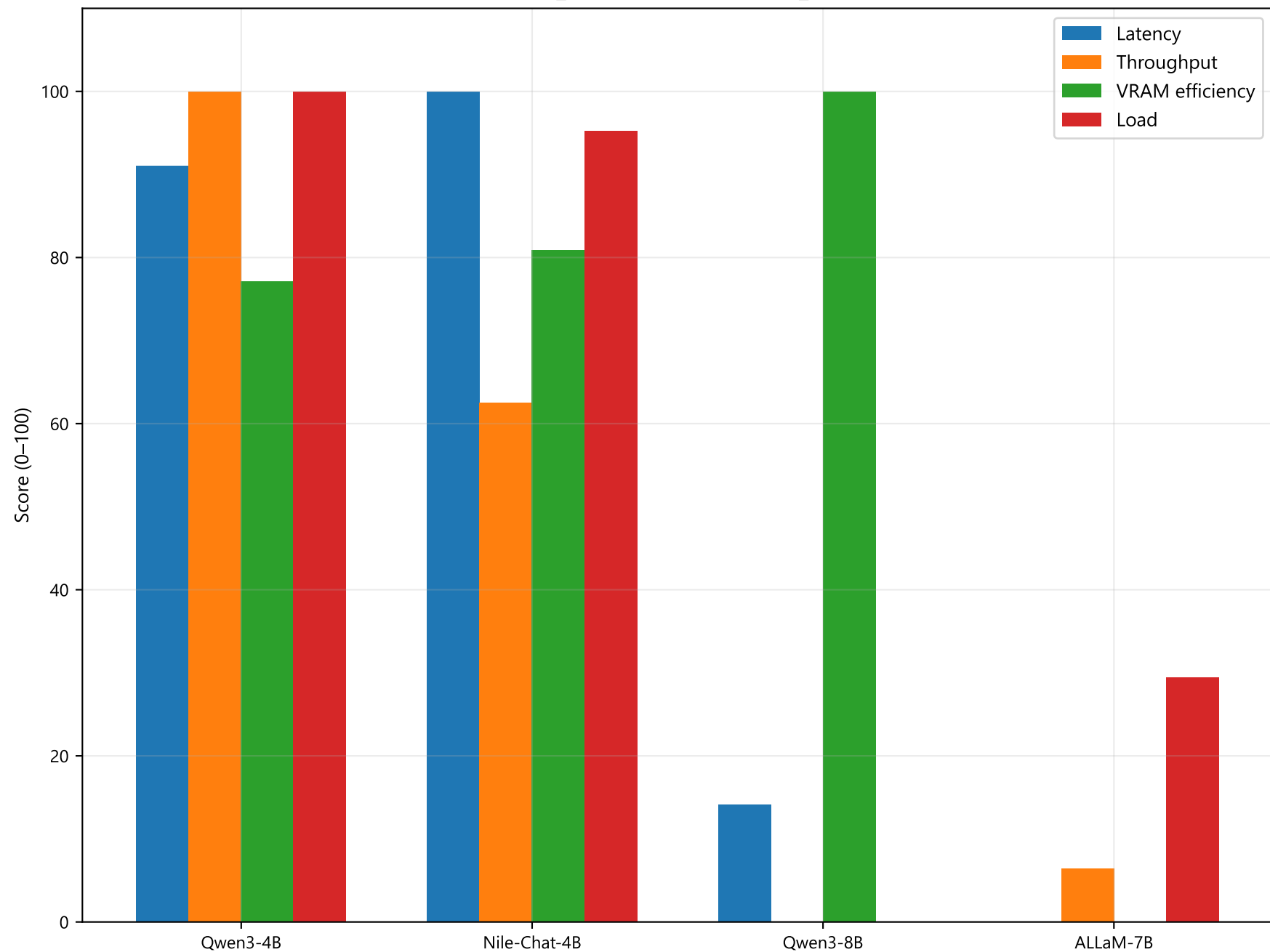


Cold load time



Normalized score breakdown (OK models)

Source: llm_leaderboard.csv · score_* columns



Leaderboard & timing detail

Robot realtime leaderboard

Rank	Model	Robot	TTFT (s)	tok/s	VRAM pk	Mode
1	Qwen3-4B	92.53	0.618	11.98	8430	bf16
2	Nile-Chat-4B	85.42	0.538	8.90	8148	bf16
3	Qwen3-8B	21.36	1.303	3.76	6722	int4
4	ALLaM-7B	4.88	1.429	4.29	14194	bf16

Timing & resources (A001)

Model	Generate (s)	Tok	Chars/s	Peak RAM	Model VRAM
Qwen3-4B	2.59	31	31.3	5572	7878
Qwen3-8B	2.13	8	8.5	5255	6170
ALLaM-7B	2.10	9	14.3	5124	13642
Nile-Chat-4B	2.70	24	26.7	5003	7596

Sample responses — A001: هدراهنل هي لماع

Auto quality scores 5/5 for OK models — manual Arabic review still required.

Qwen3-4B-Instruct-2507 | TTFT 0.618s | 11.98 tok/s | pass 5/5

هديرت اذام. شادحل نع تامولعم وأ، ندمل نع تامولعم، تامولعم لدابت لماع هدراهنل

Qwen3-8B | TTFT 1.303s | 3.76 tok/s | pass 5/5

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ALLaM-7B | TTFT 1.429s | 4.29 tok/s | pass 5/5

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Nile-Chat-4B | TTFT 0.538s | 8.90 tok/s | pass 5/5

هدراهنل هي نع فرعت زيع. اهجات حم ةجاح ي ي ف ك دعاسأ ناشع انه ان! ال ه

Failed models (no response):

- Gemma3-4B-IT: gated HF auth required
- Gemma3-12B-IT: gated HF auth required
- Jais-2-8B: gated HF auth required